Calculus 2 Sketching Part A Curves

2

2unit

Selective

Q 1 ACE

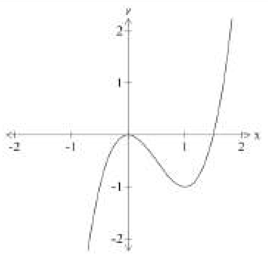
A function $f(x)$ is defined by $f(x) = 2x^3 - 3x^2$.

(i) Find all solutions for $f(x) = 0$. 1

(ii) Find any turning points and determine their nature. 3

(iii) Sketch the curve and indicate where the curve cuts the x-axis. 2

(iv) Find the values of x for which $f(x) < 0$ 1

13(c) (i)	$2x^3 - 3x^2 = 0$ $x^2(2x-3) = 0$ $x = 0, x = \frac{3}{2}$	1 Mark: Correct answer.
13(c) (ii)	$f(x) = 2x^3 - 3x^2$ Turning points $f'(x) = 0$ $f'(x) = 6x^2 - 6x$ $6x^2 - 6x = 0$ $f''(x) = 12x - 6$ $6x(x-1) = 0$ $x = 0, x = 1$ When $x = 0$, $y = 0$ then $f''(x) = -6 < 0$ Maxima. When $x = 1$, $y = -1$ then $f''(x) = 6 > 0$ Minima. Maximum turning point at $(0, 0)$. Minimum turning point at $(1, -1)$.	3 Marks: Correct answer. 2 Marks: Finds the turning points. 1 Mark: Obtains the first derivative and recognises $f'(x) = 0$.
13(c) (iii)		2 Marks: Correct answer. 1 Mark: Makes some progress towards sketching the curve.
13(c) (iv)	$f(x) < 0$ when $x < 0$ or $0 < x < \frac{3}{2}$	1 Mark: Correct answer.

Q2 GIRRAWEEN

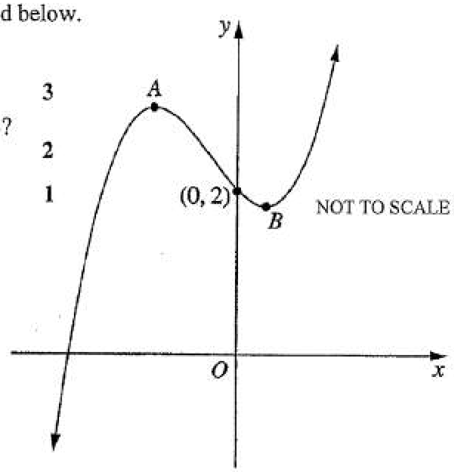
The graph of $y = x^3 + x^2 - x + 2$ is sketched below.

The points A and B are the turning points.

(i) Find the coordinates of A and B. 3

(ii) For what values of x is the curve concave up? Give reasons for your answer. 2

(iii) For what values of k has the equation $x^3 + x^2 - x + 2 = k$ three real solutions? 1



(4)(d) $y = x^3 + x^2 - x + 2$

$\frac{dy}{dx} = 3x^2 + 2x - 1$

A and B are stationary points, so need to solve $\frac{dy}{dx} = 0$:

$3x^2 + 2x - 1 = 0$

$3x^2 + 3x - x - 1 = 0$

$3x(x+1) - 1(x+1) = 0$

$(3x-1)(x+1) = 0$

$x = \frac{1}{3}$ or $x = -1$

when $x = -1$: $y = (-1)^3 + (-1)^2 - (-1) + 2$
$= -1 + 1 + 1 + 2$
$y = 3$

when $x = \frac{1}{3}$: $y = (\frac{1}{3})^3 + (\frac{1}{3})^2 - (\frac{1}{3}) + 2$
$= \frac{49}{27}$
$y = 1\frac{22}{27}$

$\therefore A(-1, 3)$, $B(\frac{1}{3}, 1\frac{22}{27})$ (3m)

(4)(d)(ii) $\frac{dy}{dx} = 3x^2 + 2x - 1$

$\frac{d^2y}{dx^2} = 6x + 2$

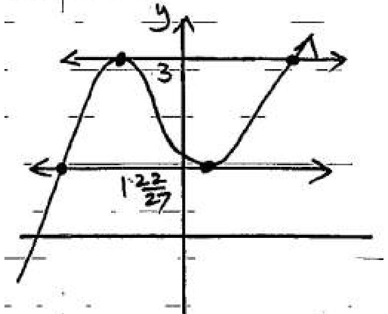
concave up when $\frac{d^2y}{dx^2} > 0$

i.e. when $6x + 2 > 0$

$6x > -2$

$\therefore$ concave up when $x > -\frac{1}{3}$ (2m)

(4)(d)(iii)



$x^3 + x^2 - x + 2 = k$ represents the intersection of $y = x^3 + x^2 - x + 2$ and $y = k$.

There will be 3 real solutions for

$1\frac{22}{27} < k < 3$ (1m)

(Note: only 2 roots if $k = 1\frac{22}{27}$ or $k = 3$)

Q 3 **ACE**

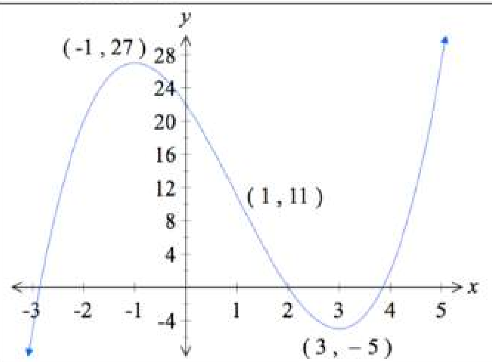
Let $f(x)=x^3-3x^2-9x+22$

(i) Find the coordinates of the stationary points and determine their nature. **3**

(ii) Find the coordinates of the point of inflexion. **2**

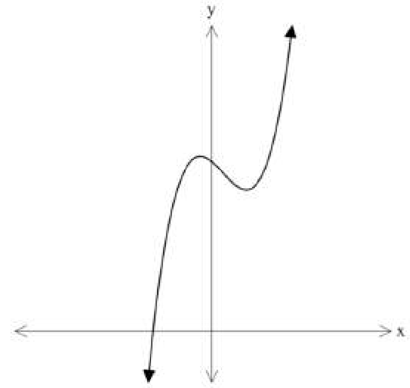
(iii) Sketch the graph of $y=f(x)$, indicating where the curve meets the y-axis, stationary points and points of inflexion. **2**

(iv) For what values of x is the graph of $y=f(x)$ concave down? **1**

13(b) (i)	$f(x)=x^3-3x^2-9x+22$ Stationary points $f'(x)=0$ $f'(x)=3x^2-6x-9$ $\quad 3(x^2-2x-3)=0$ $=3(x^2-2x-3)$ $\quad 3(x-3)(x+1)=0$ $f''(x)=6x-6$ $\quad x=-1,\ x=3$ When $x=-1, y=27$ then $f''(x)=-12<0$ Maxima. When $x=3, y=-5$ then $f''(x)=12>0$ Minima. Maximum turning point at $(-1,27)$. Minimum turning point at $(3,-5)$.	3 Marks: Correct answer. 2 Marks: Finds the stationary points. 1 Mark: Correct differentiation to determine the stationary points.
13(b) (ii)	Possible points of inflexion $f''(x)=0$ $\quad 6x-6=0$ $6(x-1)=0$ $x=1$ When $x=1, y=11$ Check for change in concavity When $x=0.9$ then $f''(x)=6\times0.9-6<0$ When $x=1.1$ then $f''(x)=6\times1.1-6>0$ Hence $(1,11)$ is a point of inflexion.	2 Marks: Correct answer. 1 Mark: Finds the point of inflexion.
13(b) (iii)		2 Marks: Correct answer. 1 Mark: Correct shape or shows some understanding.
13(b) (iv)	Function is concave down when $x<1$ (from the graph)	1 Mark: Correct answer.

Q 4 **ASCHAM**

The graph of $y=x^3-x^2-x+6$ is sketched below.



Not to scale

i) Find the coordinates of the stationary points. **(3)**

ii) Find any point(s) of inflexion **(2)**

iii) For what values of x is the curve decreasing and concave up? **(1)**

iv) For what values of p has the equation

$x^3-x^2-x+6=p$ exactly two real solutions? **(2)**

a) $y=x^3-x^2-x+6$

i) For stationary points $y'=0$

$y'=3x^2-2x-1$

$(3x+1)(x-1)=0$

$x=-\frac{1}{3},\ x=1$

$y=6\frac{5}{27}\quad y=5$

$\therefore$ stationary points are

$(-\frac{1}{3},6\frac{5}{27}),\ (1,5)$ 3

$\frac{167}{27}$

ii) For points of inflexion $y''=0$

$6x-2=0$

$x=\frac{1}{3}$

$y=5\frac{16}{27}=\frac{151}{27}$ ✓

$\therefore (\frac{1}{3},5\frac{16}{27})$ is a possible point of inflexion

Check concavity

x	0	$\frac{1}{3}$	1
y''	-2	0	4

✓

$\therefore$ change in concavity 2

$\therefore (\frac{1}{3},5\frac{16}{27})$ is a point of inflexion

[if state there is change in concavity on diagram, that is OK for the mark]

iii) $\frac{1}{3}<x<1$ 1

iv) $p=5$ and $p=6\frac{5}{27}$ 2

Q5 BARKER

For the curve $f(x)=\dfrac{2}{3}x^3-8x+1$,

(i) Find the coordinates of the stationary point(s) and determine their nature. **3**
(ii) Determine any point(s) of inflection. **2**
(iii) Sketch the curve, showing all of the above features. **2**
(iv) State the domain of x for which $f(x)$ is monotonic increasing. **1**

d) $f(x) = \frac{2}{3}x^3 - 8x + 1$

(i) $f'(x) = 2x^2 - 8$

stationary points when $f'(x) = 0$

$0 = 2x^2 - 8$

$= x^2 - 4$

$= (x-2)(x+2)$

$\therefore x = \pm 2$.

x	-3	-2	0	2	3
$f'(x)$	10	0	-8	0	10
	/	—	\	—	/

$\therefore \left(-2, \frac{35}{3}\right)$ max T.P.

$\left(2, -\frac{29}{3}\right)$ min T.P.

OR

$f''(x) = 4x$

when $x=-2$ $f''(x) = -8$ $\therefore$ concave down $\cap$ max T.P

when $x = 2$ $f''(x) = 8$ $\therefore$ concave up $\cup$ min T.P.

$\left(-2, \frac{35}{3}\right)$ max TP

$\left(2, -\frac{29}{3}\right)$ min T.P.

d) (ii) Possible POI when $f''(x) = 0$

$f''(x) = 4x$

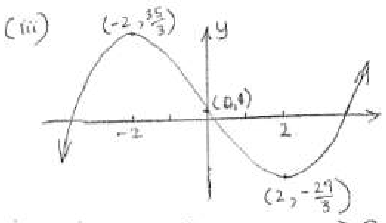
$4x = 0$

$x = 0$

x	-1	0	1
$f''(x)$	-4	0	4
	$\cap$		$\cup$

$\therefore$ $(0,1)$ is a point of inflexion.

(iii)



(iv) $x < -2$ or $x > 2$

Q6 FRENSHAM

For the curve $y = x^3(3-x)$

(i) Find any stationary points and determine their nature. **3**

(ii) Draw a sketch of the curve showing the stationary points, inflexion points and intercepts on the axes. **3**

(b)
(i)

$y = x^3(3-x) = 3x^3 - x^4$

$y' = 9x^2 - 4x^3$

Stationary points where $y' = 0$

$9x^2 - 4x^3 = 0$

$x^2(9-4x) = 0$

$x = 0$ or $x = \frac{9}{4}$

$y = 0$ or $y = 8.543$

$y'' = 18x - 12x^2$

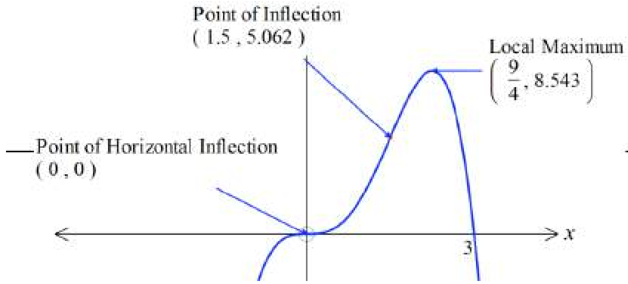
$x = 0,\ y'' = 0$ so possible inflexion

test $x = -1,\ y'' = ,-30$ $x = 1$ $y'' = 6$ so change of concavity

so (0,0) is horizontal inflexion

$x = \frac{9}{4},\ y'' = -20\frac{1}{4}$ $\therefore$ concave down

so $\left(\frac{9}{4}, 8.543\right)$ is a local maximum.



(ii) Use second derivative to check for other turning points.

$y'' = 18x - 12x^2$

$y'' = 0$ when $18x - 12x^2 = 0$

$6x(3-2x) = 0$

$x = 0$ or $x = \frac{3}{2}$

$x = 0$ is horizontal inflexion found in part i)

$x = \frac{3}{2},\ y = 5\frac{1}{16}$

$x = 2,\ y'' = -12$

$x = 1,\ y'' = 6$

$\therefore$ change of concavity so inflexion at $\left(\frac{3}{2}, 5\frac{1}{16}\right)$

Intercepts on x axis $x^3(3-x) = 0$

(ii) $x = 0$ or $x = 3$

Q7 BAULKHAM

For the function $y = x^3 - 6x^2 + 9x + 1$ find the

(i) Stationary points and determine their nature. **3**

(ii) Coordinates of any point of inflexion. **2**

(iii) Hence sketch the curve for $-1 \le x \le 5$ **3**

$y = x^3 - 6x^2 + 9x + 1$

$y' = 3x^2 - 12x + 9$

For stat pts $y' = 0$

$3(x^2 - 4x + 3) = 0$

$(x-3)(x-1) = 0$

$x = 3, 1$
$y = 1, 5$ — ①

coordinates of stats pts are $(3,1), (1,5)$

$y'' = 6x - 12$

at $x = 1$

$y'' = 6 - 12 = -6 < 0$ — ①

$\therefore$ max at $(1,5)$

at $x = 3$

$y'' = 18 - 12 = 6 > 0$ — ①

$\therefore$ min at $(3,1)$

(ii) For P.O.I

$y'' = \frac{d^2y}{dx^2} = 0$

$6x - 12 = 0$

$x = 2$ — ①

x	1	2	3
y''	-6	0	6

— ①

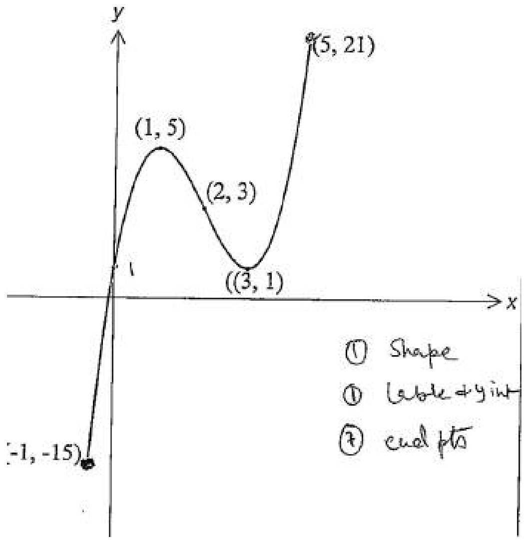
$\therefore$ concavity changes. Hence $(2,3)$ is a P.O.I

(iii) For $-1 \le x \le 5$

when $x = -1$, $y = -15$

$x = 5 \quad y = 21$

$\therefore$ end pts are $(-1,-15)$ $(5,21)$



① Shape
① lable + y int
① end pts

Q8 BAULKHAM

Consider the function given by $f(x) = 2x^3 - 3x^2 - 36x + 26$.

(i) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. **3**

(ii) Find the coordinates of any point of inflection. **2**

(iii) Hence sketch the graph of $f(x) = 2x^3 - 3x^2 - 36x + 26$ by showing the above information. **2**

(iv) For what values of x is the curve concave down and decreasing? **2**

Q13a $f(x) = 2x^3 - 3x^2 - 36x + 26$

$f'(x) = 6x^2 - 6x - 36$

For stat pts $6x^2 - 6x - 36 = 0$

$x^2 - x - 6 = 0$

$x = 3, -2$

$\therefore$ coordinates of stat pts are $(3,-55)$ & $(-2,70)$

$\frac{d^2y}{dx^2} = f''(x) = 12x - 6$

at $x = 3$ $\quad \frac{d^2y}{dx^2} = 36 - 6 = 30 > 0$ ✓ $\therefore$ min at $(3,-55)$

at $x = -2$

$\frac{d^2x}{dy^2} = -24 - 6 = -30 < 0$ $\therefore$ max at $(-2,70)$ ✓

(ii) For possible P.O.I $\frac{d^2y}{dx^2} = 0$ ✓

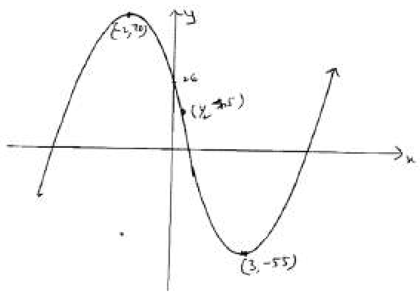
$12x - 6 = 0 \Rightarrow x = \frac{1}{2}$

x	-1	$\frac{1}{2}$	1
$\frac{d^2y}{dx^2}$	-18	0	6

$\therefore$ There is change in concavity at $x = \frac{1}{2}$

$\therefore (\frac{1}{2}, 7.5)$ is P.O.I ✓

(must show change in concavity)



(iv) for decreasing fn then $f'(x) < 0$

$6x^2 - 6x - 36 < 0$

$(x-3)(x+2) < 0$

$-2 < x < 3$ — ✓

For concave down $12x - 6 < 0$

$x < \frac{1}{2}$

$\therefore -2 < x < \frac{1}{2}$ — ✓

Q9 GOSFORD

A function is given by $f(x) = -x^3 + \frac{3x^2}{2} + 6x$

(i) Find the coordinates of the stationary points of $f(x)$ and determine their nature. **3**

(ii) Find the coordinates of the point of inflexion of $f(x)$. **1**

(iii) Hence, sketch the graph $y = f(x)$ showing the stationary points, point of inflexion and y intercept. **3**

(iv) For what values of x is the function decreasing? **1**

13.a (i) $f(x) = -x^3 + \frac{3x^2}{2} + 6x$

$f'(x) = -3x^2 + 3x + 6$

$f''(x) = -6x + 3$

Stationary points $f'(x) = 0$

$0 = -3x^2 + 3x + 6$

$0 = x^2 - x - 2$

$0 = (x-2)(x+1)$

$\therefore x = -1, 2$

At $x = -1$, $y = -(-1)^3 + \frac{3(-1)^2}{2} + 6(-1)$

$= -3.5$

$f''(x) = -6(-1) + 3$

> 0

$\therefore (-1, -3.5)$ is a local minimum.

At $x = 2$, $y = -(2)^3 + \frac{3(2)^2}{2} + 6(2)$

$= 10$

$f''(x) = -6(2) + 3$

< 0

$\therefore (2, 10)$ is a local maximum

(ii) Point of inflexion $f''(x) = 0$

$0 = -6(x) + 3$

$-3 = -6x$

$x = \frac{1}{2}$

$y = -(\frac{1}{2})^3 + \frac{3(\frac{1}{2})^2}{2} + 6(\frac{1}{2})$

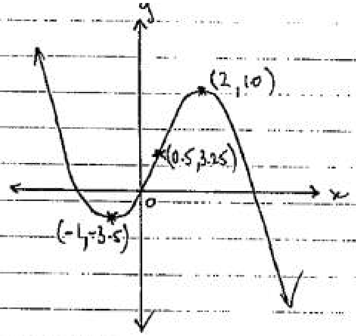
$= 3.25$

$(0.5, 3.25)$

Check:

x	$\frac{1}{2}^-$	$\frac{1}{2}$	$\frac{1}{2}^+$
$f''(x)$	$+$	0	$-$

(iii)



(iv) $x < -1$ and $x > 2$

Q10 HURLSTONE

A function $f(x)$ is defined by $f(x) = x^3 - 3x^2$.

(i) Find all values of x that satisfy $f(x) = 0$. **1**

(ii) Find the coordinates of the stationary points and determine their nature. **3**

(iii) Find the coordinates of any points of inflexion, justifying your answer. **2**

(iv) Sketch the curve $y = f(x)$ showing all important features. **2**

(v) For what values of x is the graph of $y = f(x)$ concave down? **1**

) (i) P5

$f(x) = 0$

$x^3 - 3x^2 = 0$

$x^2(x-3) = 0$

$\therefore x = 0$ or 3.

(ii) H6

$f'(x) = 3x^2 - 6x$

$f''(x) = 6x - 6$

Possible stationary points occur @ $f'(x) = 0$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$\therefore x = 0$ or 2

Test $x = 0$, $f''(0) = -6 < 0$

$\therefore$ Relative maximum @ $(0,0)$

Test $x = 2$, $f''(2) = 6 > 0$

$\therefore$ Relative minimum @ $(2,-4)$

(iii) H6

$f''(x) = 0$

$6x - 6 = 0$

$\therefore x = 1$

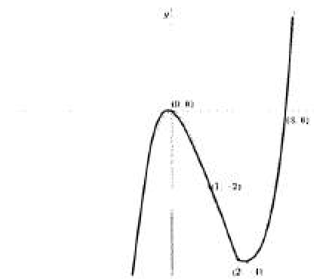
Test $x = 0$, $f''(0) = -6$

Test $x = 2$, $f''(2) = 12$

$\therefore$ Change in concavity either side of $x = 1$

$\therefore$ Point of inflexion @ $(1,-2)$

(iv) H6



(v) H6

Concave down $f''(x) < 0$

$6x - 6 < 0$

$\therefore x < 1$

Q11 HORNSBY GIRLS

Consider the curve $y = x^3 + 4x^2 - 3x + 2$.

(i) Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. **2**

(ii) Find the coordinates of the stationary points and determine their nature. **2**

(iii) Find the coordinates of any points of inflexion. **2**

(iv) Sketch the graph of $y = x^3 + 4x^2 - 3x + 2$, clearly showing any stationary points, points of inflexion and the y-intercept. **2**

(v) Hence find the number of solutions to the equation $x^3 + 4x^2 - 3x + 2 = -1$. **1**

(a)
(i)
$y = x^3 + 4x^2 - 3x + 2$
$\dfrac{dy}{dx} = 3x^2 + 8x - 3$
$\dfrac{d^2y}{dx^2} = 6x + 8$

① for $\dfrac{dy}{dx}$
① for $\dfrac{d^2y}{dx^2}$

(ii)
Let $\dfrac{dy}{dx} = 0$
$3x^2 + 8x - 3 = 0$
$3x^2 + 9x - x - 3 = 0$
$3x(x+3) - 1(x+3) = 0$
$(3x-1)(x+3) = 0$
$x = -3,\ x = \dfrac{1}{3}$

When $x = -3$
$y = (-3)^3 + 4(-3)^2 - 3(-3) + 2$
$= 20$
When $x = \dfrac{1}{3}$
$y = \left(\dfrac{-1}{3}\right)^3 + 4\left(\dfrac{-1}{3}\right)^2 - 3\left(\dfrac{-1}{3}\right) + 2$
$= \dfrac{40}{27} (= 1.4)$

Testing:
When $x = -3$
$\dfrac{d^2y}{dx^2} = 6(-3) + 8$
$= -10 < 0$
Therefore maximum at $(-3, 20)$
When $x = \dfrac{1}{3}$
$\dfrac{d^2y}{dx^2} = 6(\dfrac{1}{3}) + 8$
$= 10 > 0$
Therefore minimum at $\left(\dfrac{1}{3}, \dfrac{40}{27}\right)$

① for minimum stat. pt
① for maximum stat. pt

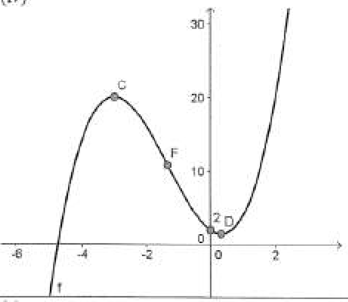
Students are reminded that if they are using a table of values to determine the nature of the stationary point, NOT to just use + and – signs in the table.

(iii)
Let $\dfrac{d^2y}{dx^2} = 0$
$6x + 8 = 0$
$x = \dfrac{-4}{3}$
When $x = \dfrac{-4}{3}$
$y = \left(\dfrac{-4}{3}\right)^3 + 4\left(\dfrac{-4}{3}\right)^2 - 4\left(\dfrac{-4}{3}\right) + 2$
$= \dfrac{290}{27} (\approx 10.74..)$
Test
$x = -2,\ y'' = -4$
$x = -1,\ y'' = 2$
Therefore, change in concavity at $\left(\dfrac{-4}{3}, \dfrac{290}{27}\right)$

① for point of inflexion
① for testing change of concavity

Some students did not test for a change in concavity.

(iv)



① for shape
① for labelling (especially y intercept)

It was disappointing to see so many poorly drawn graphs.
- *Do NOT use a feathered line.*
- *Your graph should be a smooth, freehand CURVE.*
- *Show that the graph cuts the x axis.*
- *Label key points but no need to annotate with "maximum", "minimum", etc.*

(v)
From inspection, there is one solution to the equation
$x^3 + 4x^2 - 3x + 2 = -1$

Q12 INDEPENDENT

A function is given by $f(x)=12x-3x^2-2x^3$

(i) Find the coordinates of the stationary points of $f(x)$ and determine their nature. **3**

(ii) Hence sketch the graph $y=f(x)$ showing the stationary points and y–intercept. **2**

(iii) For what values of x is the function decreasing? **1**

(iv) For what values of k will $12x-3x^2-2x^3+k=0$ have 2 real solutions? **2**

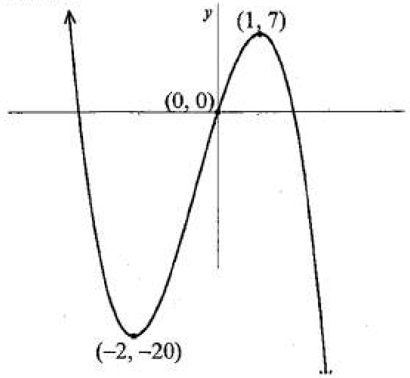
a)(i) Outcome Assessed: H6 M:

Answer:

$$f(x)=12x-3x^2-2x^3$$
$$f'(x)=12-6x-6x^2$$
$$0=12-6x-6x^2$$
$$0=x^2+x-2$$
$$0=(x+2)(x-1)$$
$$x=-2,1$$
$$f(-2)=-24-12+16$$
$$=-20$$
$$f''(x)=-6-12x$$
$$f''(-2)=-6+24$$
$$=18>0 \text{ concave up}$$
$$\therefore(-2,-20) \text{ is a minimum}$$
$$f(1)=12-3-2$$
$$=7$$
$$f''(1)=-18<0 \text{ concave down}$$
$$\therefore(1,7) \text{ is a maximum}$$

a)(ii) Outcome Assessed: H6

Answer:



a)(iii) Outcome Assessed: H6

Answer:

$x<-2$ and $x>1$

a)(iv) Outcome Assessed: H6

Answer:

Move function down 7, $k=-7$ or up 20, $k=20$

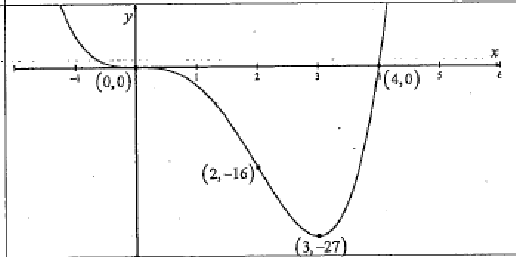
Q13 INDEPENDENT

Consider the function $f(x)=x^4-4x^3$.

(i) Find the stationary points on the curve and determine their nature. **4**

(ii) Find any points of inflexion. **2**

(iii) Sketch the curve $f(x)=x^4-4x^3$ showing all important features including the intercepts. **2**

c)(i)	$f(x)=x^4-4x^3$ $f'(x)=4x^3-12x^2$ stationary points when $f'(x)=0$ $4x^3-12x^2=0$ $4x^2(x-3)=0$ $x=0,3$	1 for values of x
	$f(0)=0$ $f(3)=3^4-4(3)^3$ $=-27$	1 for y values
	$SP=(0,0)\ (3,-27)$ test stationary points $f''(x)=12x^2-24x$ $f''(0)=0$ test concavity (table below) change in concavity, $\therefore (0,0)$ horizontal point of inflexion	1 correct identification of horizontal P.O.I
	$f''(3)=12(3)^2-24(3)$ $=36>0$ concave up $\therefore$ minimum at $(3,-27)$	1 correct identification of minimum
c)(ii)	Possible points of inflexion occur when $f''(x)=0$ $12x^2-24x=0$ $12x(x-2)=0$ $x=0,2$	1 for x values.
	$f(2)=2^4-4(2)^3$ $=-16$ (table below) Change in concavity $\therefore (2,-16)$ is a P.O.I	1 for correct test & conclusion. (Origin already tested)
c)(iii)		1 stationary points and shape for their information 1 correct intercepts and their points of inflexion

x	0^-	0	0^+
$f''(x)$	+ve	0	-ve

x	2^-	2	2^+
$f''(x)$	-ve	0	+ve

Q14 INDEPENDENT

Consider the curve $y = x^3 - 12x^2 + 36x$.

(i) Find the x and y intercepts. **2**

(ii) Find any stationary points and determine their nature. **3**

(iii) Hence sketch for $-1 \le x \le 7$, showing all features found in parts (i) and (ii). **2**

(i) y intercept $= 0$.

x intercepts.

$y = 0$

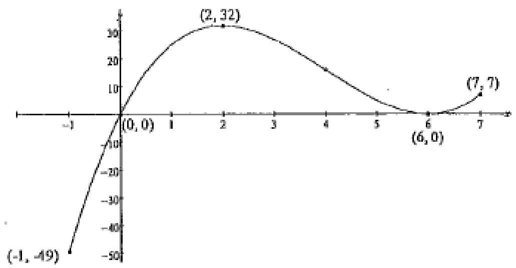
$0 = x^3 - 12x^2 + 36x$

$0 = x\left(x^2 - 12x + 36\right)$

$0 = x(x-6)^2$

$x = 0,\ 6$

(iii)



(ii) $y = x^3 - 12x^2 + 36x$

$\dfrac{dy}{dx} = 3x^2 - 24x + 36$

Stationary points occur when $\dfrac{dy}{dx} = 0$

$3x^2 - 24x + 36 = 0$

$x^2 - 8x + 12 = 0$

$(x-2)(x-6) = 0$

$x = 2,\ 6$

Stationary Points at $(2,\ 32)$ and $(6,\ 0)$

Test stationary points.

$\dfrac{d^2y}{dx^2} = 6x - 24$

At $(2,\ 32)$, $\dfrac{d^2y}{dx^2} = -12 < 0$ $\therefore$ a maximum turning point

At $(6,\ 0)$, $\dfrac{d^2y}{dx^2} = 12 > 0$ $\therefore$ a minimum turning point

Q15 INDEPENDENT

A function is defined as $f(x) = x^3 - 3x^2$.

(i) Find the coordinates of the stationary points and determine their nature. **3**

(ii) Find the coordinates of the point of inflexion. **1**

(iii) Sketch the graph of $y = f(x)$ indicating clearly the stationary points, the point of inflexion and the x-intercepts. **3**

(iv) Find the minimum value of the function in the interval $-2 \le x \le 3$. **1**

(a)(i)

$f(x) = x^3 - 3x^2$

$f'(x) = 3x^2 - 6x = 0$ for stationary points (1 mark)

$x = 0, 2$

$f''(x) = 6x - 6 = 0$ for points of inflexion

$x = 1$

At $x = 0$, $f(x) = 0$, $f''(x) = -6$ (concave down) $\therefore (0.0)$ is a Maximum T.P.

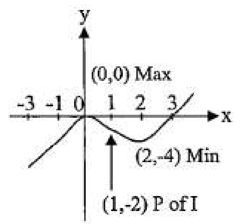
Similarly $(2,-4)$ is a Minimum T.P. (1 mark + 1 mark)

(a)(ii)

At $x = 1$, $f''\left(1^+\right) > 0$, $f''\left(1^-\right) < 0$, $f''(x) = 0$, and changes sign. $f(1) = -2$

(1,-2) is a point of inflexion (1 mark)

(a)(iii)



NOT TO SCALE

(1 mark each for stat.pts, POI and x-intercepts done correctly)

(a)(iv)

From the graph, minimum occurs at either x = -2 or x = 2

At $x = 2$, $f(2) = -4$, $f(-2) = -20$

Minimum in the given interval is -20 (1 mark)

Q16 INDEPENDENT

A function is defined by $f\left(x\right)=x^3-3x^2-9x+22$.

(i) Find the coordinates of the turning points of the graph $y=f\left(x\right)$, and determine their nature. **3**

(ii) Find the coordinates of the point of inflexion. **2**

(iii) Hence sketch the graph of $y=f\left(x\right)$, showing the turning points, the point of inflexion and where the curve meets the y–axis. **2**

(iv) For what values of x is the graph of $y=f\left(x\right)$ concave up? **1**

Sample Answer:-

$f\left(x\right)=x^3-3x^2-9x+22$

$f'\left(x\right)=3x^2-6x-9$

Stationary points occur when $f'\left(x\right)=0$

$0=3x^2-6x-9$

$0=x^2-2x-3$

$0=\left(x-3\right)\left(x+1\right)$

$x=-1,\ 3$

$f\left(-1\right)=-1-3+9+22$

$=27$

$f\left(3\right)=27-27-27+22$

$=-5$

Turning points $\left(-1,\ 27\right),\ \left(3,\ -5\right)$

Test nature of points.

$f''\left(x\right)=6x-6$

Test $\left(-1,\ 27\right)$

$f''\left(-1\right)=-12<0$ $\therefore$ concave down. $\left(-1,\ 27\right)$ is a maximum

Test $\left(3,\ -5\right)$

$f''\left(3\right)=12>0$ $\therefore$ concave up. $\left(3,\ -5\right)$ is a minimum.

Sample Answer:-

$f''\left(x\right)=6x-6$

Possible points of inflexion occur when $f''\left(x\right)=0$

$6x-6=0$

$6x=6$

$x=1$

$f\left(1\right)=1-3-9+22$

$=11$

Possible point of inflexion $\left(1,\ 11\right)$

Test point of inflexion.

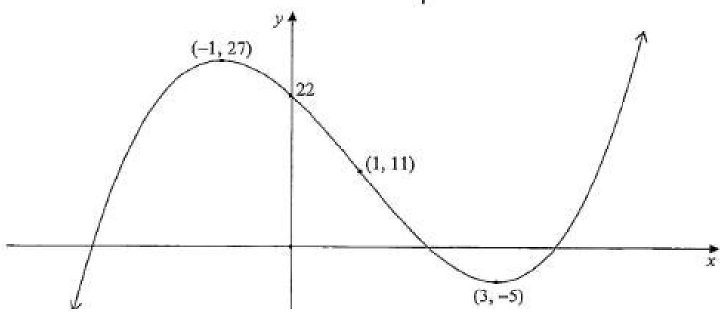
x	-1	1	3
$f''\left(x\right)$	-12	0	12

Change in concavity, $\therefore$ $\left(1,\ 11\right)$ is a point of inflexion.

a) (iv) 1 marks

Sample Answer:-

$x>1$



Q17 INDEPENDENT

A function $f(x)$ is defined by $f(x)=x\left(x^2-3x-9\right)$.

(i) Find the turning points for the curve $y=f(x)$ and determine their nature. **3**

(ii) Find the coordinates of the point of inflexion. **1**

(iii) Sketch the graph of $y=f(x)$ showing the turning points and point of inflexion. **2**

(vi) Find the values of x for which both $f'(x)<0$ and $f''(x)>0$. **2**

(i) $f'(x)=3x^2-6x-9$

$3(x-3)(x+1)=0$

$x=3,-1$

$f''(x)=6x-6.=0$

$x=1$

At $x=3$, $y=-27$

$f''(x)>0$ ∪ Conc up.

$\therefore$ $(3,-27)$ Min TP.

Simi $(-1,5)$ Max T.P.

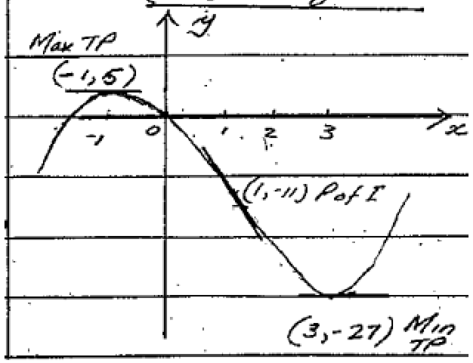
(ii) At $x=1$ $y=-11$

$x>1$ $f''(x)>0$

$x<1$ $f''(x)<0$

$\therefore$ Conc changes

$(1,-11)$ Pt of infl.



$f'(x)<0$ $-1<x<3$

$f'(x)>0$ $x>1$

$\therefore$ Both T $1<x<3$

Q18 **IGS**

A function $f(x)$ is defined by $f(x)=x^2(3-x)$.

(i) Find the stationary points for the curve $y=f(x)$ and determine their nature. 3

(ii) Sketch the graph of $y=f(x)$ showing the stationary points and x-intercepts. 2

2. i. $f(x) = x^2(3-x)$
$= 3x^2 - x^3$

For Stationary points
$f'(x) = 0$.

$f'(x) = 6x - 3x^2$

$6x - 3x^2 = 0$

$3x(2-x) = 0$

- $x=0 \to y=0$
- $x=2 \to y=4$.

$\therefore (0,0), (2,4)$

At $(0,0)$, $f''(0) = 6-6x$
$= 6 > 0$, Min

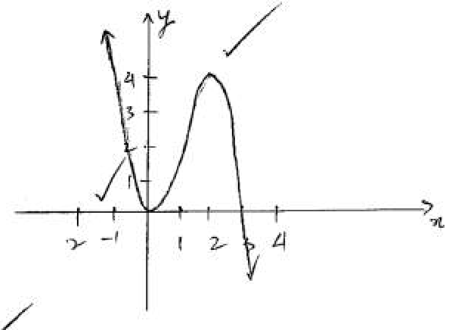
At $(2,4)$, $f''(2) = 6-12$
$= -6 < 0$, Max.

ii. x intercept

$y=0 \to 3x^2 - x^3 = 0$

$x^2(3-x) = 0$

$x=0$ and $x=3$



Q19 **JAMES RUSE**

Let $f(x) = (x+2)(x^2+4)$.

i. Show that $y=f(x)$ has no stationary points. 2

ii. Find the values of x for which $y=f(x)$ is concave up and concave down respectively. 2

iii. Sketch the graph $y=f(x)$, indicating all intercepts and any points of inflexion. 2

a) i) show $f(x)$ has no stationary points.

$f(x) = (x+2)(x^2+4)$

$f'(x) = (x+2)\cdot 2x + (x^2+4)\cdot 1$

$= 2x^2 + 4x + x^2 + 4$

$= 3x^2 + 4x + 4$

$\Delta = b^2 - 4ac$

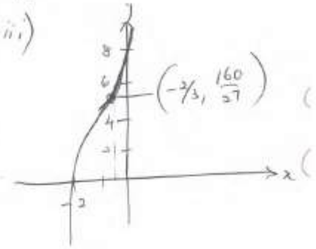
$= 16 - 4(3)(4)$

$= -32$

$f'(x) \neq 0$ since $\Delta < 0$ (no real root)

$\therefore$ no stationary points.

iii)



ii) $f''(x) = 6x + 4$

$f''(x) > 0$ concave up

$\therefore 6x + 4 > 0$

$6x > -4$

$x > -\frac{2}{3}$

1

$f''(x) < 0$ concave down

$6x + 4 < 0$

$6x < -4$

$x < -\frac{2}{3}$

1

or

x	$< -\frac{2}{3}$ (ie -0.7)	$-\frac{2}{3}$	$> -\frac{2}{3}$ (ie 0.6)
$f''(x)$	-0.2	0	0.4
	$\cap$ concave down		$\cup$ concave up

Q20 KAMBALA

For the function $y = x^3 - 3x^2 - 9x + 6$:

(i) Find the co-ordinates of any stationary points and determine their nature. **3**
(ii) Find the co-ordinates of any points of inflection. **2**
(iii) At what point does the curve cut the y-axis? **1**
(iv) For what values of x is the curve concave up? **1**
(v) Sketch the curve, showing all essential features. **2**

$y = x^3 - 3x^2 - 9x + 6$

(i) $y' = 3x^2 - 6x - 9$

For a stat. pt., $y' = 0$

$0 = 3x^2 - 6x - 9$

$0 = x^2 - 2x - 3$

$0 = (x-3)(x+1)$

$\therefore x = 3, -1$

when $x = 3$, $y = \cancel{27} - \cancel{27} - 27 + 6$, $y = -21$

when $x = -1$, $y = -1 - 3 + 9 + 6$, $y = 11$

$\therefore \boxed{\text{stat pts are } (3, -21), (-1, 11)}$

$y'' = 6x - 6$

when $x = 3$, $y'' = 18 - 6$, $y'' = 12 > 0$ $\therefore$ $(3, -21)$ is a MIN TURN PT

when $x = -1$, $y'' = -6 - 6$, $y'' = -12 < 0$ $\therefore$ $(-1, 11)$ is a MAX TURN. PT

(ii) For an inflection pt, $y'' = 0$ and changes sign.

$y'' = 6x - 6$

$6x - 6 = 0$

$\therefore x = 1$

when $x = 1$, $y = 1 - 3 - 9 + 6$, $y = -5$

Test

x	0	1	2
y''	-6	0	$+6$

$\therefore \boxed{\text{Inflection pt. at } (1, -5)}$

(iii) $y = x^3 - 3x^2 - 9x + 6$

when $x = 0$, $y = 6$. $\boxed{(0, 6)}$

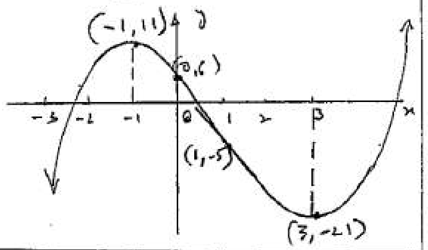
(iv) For concave up, $y'' > 0$.

$y'' = 6x - 6$

$\therefore 6x - 6 > 0$

$\therefore \boxed{x > 1}$

(v)



Q21 KILARA

Consider the curve given by $y = 1 - 12x + x^3$ for $-4 \leq x \leq 4$.

i. Find the stationary points and determine their nature. **3**

ii. Find the point of inflexion. **2**

iii. Sketch the curve for $-4 \leq x \leq 4$. **1**

(i) Given $y = 1 - 12x + x^3$

$\frac{dy}{dx} = -12x + 3x^2 = 3(x^2 - 4)$ $\qquad \frac{d^2y}{dx^2} = 6x$.

hence $\frac{dy}{dx} = 0$, $x = \pm 2$ $\quad$ ① for finding the value of x.

when $x = 2$, $y = -15$, $\frac{d^2y}{dx^2} = 12 > 0$ $\quad$ ① for determining their nature

when $x = -2$, $y = 17$, $\frac{d^2y}{dx^2} = -12 < 0$

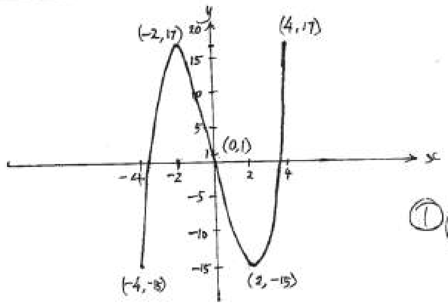
$\therefore$ stationary points are $(2, -15)$ minimum turning point, $(-2, 17)$ maximum turning point. $\quad$ ① for stationary pt

(ii) Point of inflexion at $\frac{d^2y}{dx^2} = 0$, $6x = 0$, $x = 0$, when $x = 0$, $y = 1$ $\quad$ ① for testing

x	-1	0	1
$\frac{d^2y}{dx^2}$	-6	0	6

$\therefore (0, 1)$ is the point of inflexion. $\quad$ ① for point of inflexion

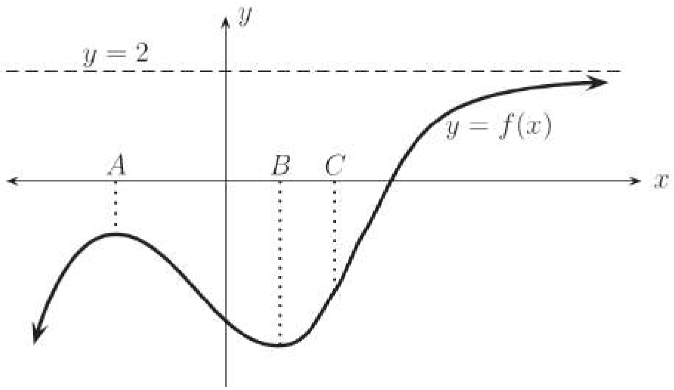
(iii)

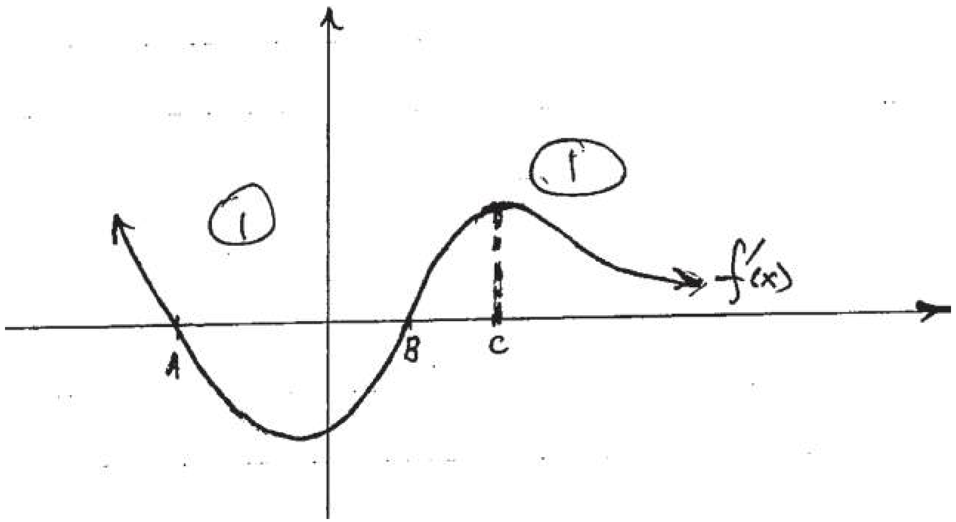


① for sketching the curve.

Q22 **KILARA**

Given the graph of $y = f(x)$ below, sketch the graph of $f'(x)$ clearly showing the points A, B and C on your graph. **2**





Q23 **KILARA**

Consider the function $f(x) = 2x^3 - 4x^2$

i) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. **3**

ii) Find the point of inflexion. **2**

iii) Sketch the curve showing where it meets the axes. **2**

$f(x) = 2x^3 - 4x^2$

$f'(x) = 6x^2 - 8x$

let $f'(x) = 0$ then

$6x^2 - 8x = 0$

$2x(3x-4) = 0$

$x = 0,\ x = \frac{4}{3}$ are stationary points

$f''(x) = 12x - 8$

$f''(0) = -8 < 0$ then maximum at $(0,0)$ (value<0 must be shown)

$f''(\frac{4}{3}) = 8 > 0$ then a minimum at $(\frac{4}{3}, -\frac{64}{27})$ (Value >0 must be shown)

Find x-intercepts

$f(x) = 0$ when $2x^3 - 4x^2 = 0$

$2x^2(x-2) = 0$

$x = 0,\ 2$

at point of inflection, $f''(x) = 0$

$f''(x) = 12x - 8 = 0$

$x = \frac{2}{3}$ is a possible pt of inflection

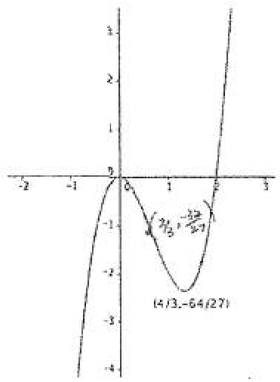
$y = \frac{-32}{27}$ (1)

must show change of concavity

x	0	$\frac{2}{3}$	1
$f''(0)$	-8	0	4
	\	—	/

(values must be shown)

Concavity changes so point of inflection at $\left(\frac{2}{3}, -\frac{32}{27}\right)$ (2)



Q24 KINCOPPAL

Consider the curve $y = x^3 - 6x^2 + 8$

(i) Find any stationary points and determine their nature. **3**

(ii) Show that there is one point of inflexion and find its co-ordinates. **2**

(iii) Hence sketch the curve for the domain $-1 \le x \le 5$ indicating important features. **2**

(iv) Find the maximum value of the function in the domain $-1 \le x \le 5$. **1**

(i)

$\frac{dy}{dx} = 3x^2 - 12x$

$\frac{dy}{dx} = 0 \Rightarrow 3x(x-4) = 0$

$\therefore x = 0, 4 \qquad y = 8, -24$

$\frac{d^2y}{dx^2} = 6x - 12$

$at\ x = 0\ \frac{d^2y}{dx^2} < 0 \qquad \Rightarrow (0,8)\ is\ \max$

$at\ x = 4\ \frac{d^2y}{dx^2} > 0 \qquad \Rightarrow (4,-24)\ is\ \min$

(ii)

$\frac{d^2y}{dx^2} = 0$

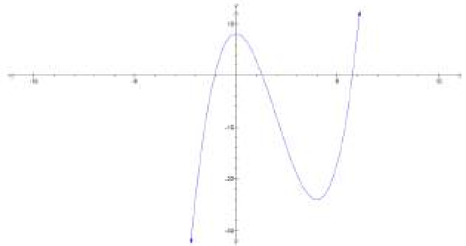
$6x - 12 = 0 \Rightarrow x = 2$

Test:

x	1	2	3
$\frac{d^2y}{dx^2}$	<0	0	>0

There is a change in concavity and hence $(2,-8)$ is a point of inflexion

(iii)



1 for endpoints: $(-1,1)$ and $(5,-17)$

1 for sketch showing important features turning point, y-intercept

(iv) Max value in the domain is 8

Q25 NSBHS

Consider the function $y = x^3 - 9x^2 + 24x$.

(i) Find all stationary points and determine their nature. **4**

(ii) Find the point of inflexion. **2**

(iii) Sketch the curve showing all important features. **2**

(i)

$y = x^3 - 9x^2 + 24x$

$y' = 3x^2 - 18x + 24$

To find stationary points,

let $y' = 0$

$3x^2 - 18x + 24 = 0$

$x^2 - 6x + 8 = 0$

$(x-2)(x-4) = 0$

$x = 2$ or $x = 4$

$y'' = 6x - 18$

When $x = 2, y'' = -6 < 0$

$\therefore$ maximum turning point at $(2,20)$

When $x = 4, y'' = 6 > 0$

$\therefore$ minimum turning point at $(4,16)$

(ii)

To find points of inflexion,

let $y'' = 0$

$6x - 18 = 0$

$x = 3$

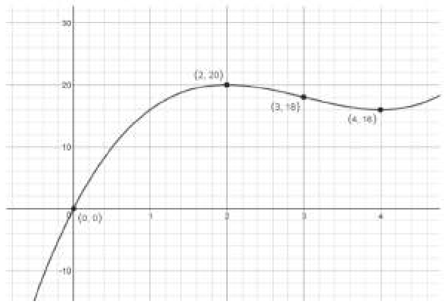
To check for change of concavity at $x = 3$,

x	2	3	4
y''	-6	0	6

$\therefore$ there is a change in concavity at $x = 3$

$\therefore (3,18)$ is a point of inflexion.

(iii)



Q26 KINGS

Consider the function $f(x) = (2-x)(x^2-4)$

(i) Find all solutions to $f(x) = 0$ **1**
(ii) Find the coordinates of the stationary points of the graph $y = f(x)$ and determine their nature. **4**
(iii) Find any points of inflection. **2**
(iv) For what values of x is the curve concave down? **1**
(v) Sketch the curve $y = (2-x)(x^2-4)$ indicating all the important features. **2**

a) i) $x = 2, -2$ (1)
ii) $f'(x) = -3x^2 + 4x + 4$ (1)
$f''(x) = -6x + 4$
Stat Pts $f'(x) = 0$
$-3x^2 + 4x + 4 = 0$
$-(3x+2)(x-2) = 0$
$x = -\frac{2}{3}, 2$ (1)

When $x = 2$, $y = 0$
$f''(2) = -8$
< 0 (1)
$\therefore (2,0)$ is a max turning pt.

When $x = -\frac{2}{3}$ $y = -9\frac{13}{27}$
$f''(-\frac{2}{3}) = 8$
> 0 (1)
$\therefore (-\frac{2}{3}, -9\frac{13}{27})$ is a min turning pt

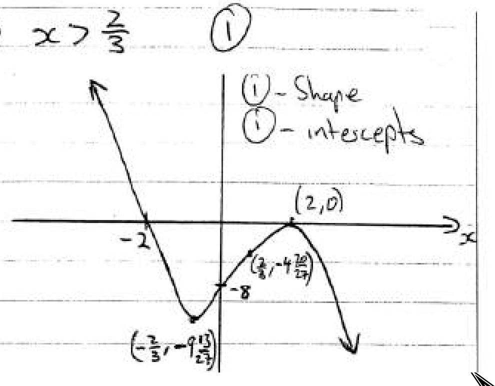
iii) Pt of inflection $f''(x) = 0$
$-6x + 4 = 0$
$x = \frac{2}{3}$ (1)
possible pt of inflection $(\frac{2}{3}, -4\frac{20}{27})$

x	$1/3$	$2/3$	1
$f''(x)$	>0	0	<0

(1)

Since concavity changes $(\frac{2}{3}, -4\frac{20}{27})$ is a pt of inflection

iv) $x > \frac{2}{3}$ (1)



Q27 KINGS

Consider the curve given by $y = 1 + 3x - x^3$, for $-2 \le x \le 3$

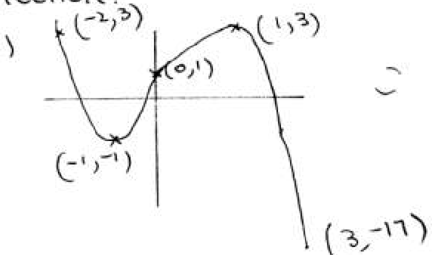
(i) Find the stationary points and determine their nature **2**
(ii) Find any points of inflection **2**
(iii) Sketch the curve for $-2 \le x \le 3$ **1**
(iv) What is the minimum value of y for $-2 \le x \le 3$ **1**

(b) $y = 1 + 3x - x^3$ $y'' = -6x$
$y' = -3x^2 + 3$ Stat pts $y' = 0$
$-3(x+1)(x-1) = 0$
$x = \pm 1$
at $x = 1$ $y = 3$
$y'' < 0$ $\therefore$ max turning pt at $(1,3)$

at $x = -1$ $y = -1$
$y'' > 0$ $\therefore$ min turning pt at $(-1,-1)$

(ii) possible pt of inflection at $y'' = 0$
$-6x = 0$
$x = 0$ $y = 1$
when $x < 0$ $y'' > 0$
$x > 0$ $y'' < 0$
i.e concavity changes
$\therefore$ at $(0,1)$ there is a pt of inflection.

iii)



(iv) min value of y for $-2 \le x \le 3$ is $y = -17$

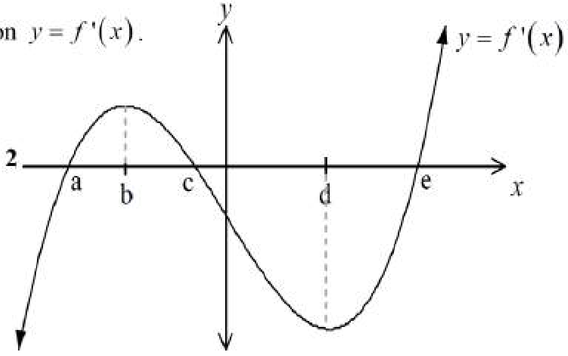
Q28 KINGS

This is the graph of the gradient function $y = f'(x)$.

(i) Copy or trace this graph and on the same axes draw a possible graph of $y = f''(x)$. **2**

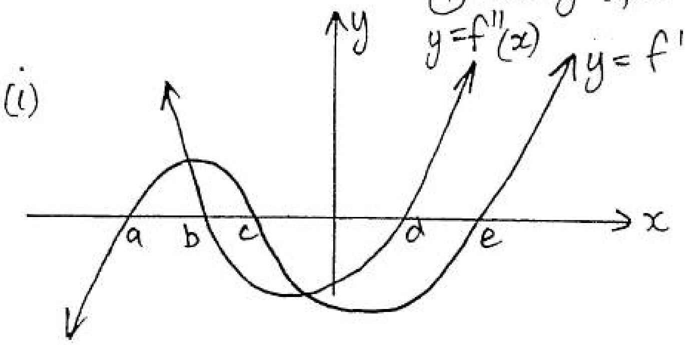
(ii) For what value(s) of x is the graph of $y = f(x)$ concave up? **2**

(You do not need to draw this graph.)



① shape
① cutting b,d.

(a)(i)



(ii) $x < b$ ①
$x > d$ ①

Q29 KINGS

A function $f(x)$ is defined by $f(x) = 7 + 4x^3 - 3x^4$.

(i) Find the coordinates of the stationary points for the curve $y = f(x)$. **2**

(ii) Determine the nature of the stationary points. **3**

(iii) Sketch the graph of $y = f(x)$ for the domain $-1 \le x \le 2$. **2**

a) i) $f(x) = 7 + 4x^3 - 3x^4$

$f'(x) = 12x^2 - 12x^3$

$f''(x) = 24x - 36x^2$

Stat pts $f'(x) = 0$

$12x^2 - 12x^3 = 0$

$12x^2(1-x) = 0$

$x = 0, \quad x = 1$

$\therefore$ Stat pts at $(0,7)$ and $(1,8)$

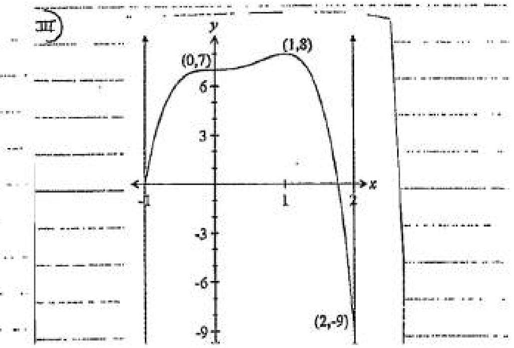
ii) At $(0,7)$ $f''(0) = 0$, $\therefore$ possible pt of inflexion

x	-0.1	0	0.1
$f''(x)$	<0	0	>0

Since concavity changes pt of inflexion

At $(1,8)$, $f''(1) < 0$ $\therefore$ maxima

iii)



Q30 KINGS

Consider the curve given by $y = x^4 - 108x + 240$.

(i) Show that there is a stationary point when $x = 3$ **2**

(ii) Using the second derivative, determine the nature of this stationary point. **2**

(iii) Show that $\frac{d^2y}{dx^2} = 0$ when $x = 0$ and explain why (0,240) is **NOT** a point of inflexion. **2**

(iv) Sketch the curve $y = x^4 - 108x + 240$ showing all the turning points.
You do not need to find the x intercepts. **2**

a) i) $y = x^4 - 108x + 240$

$\frac{dy}{dx} = 4x^3 - 108$

$= 4(x^3 - 27)$

$= 4(x-3)(x^2+3x+9)$

$\frac{dy}{dx} = 0$ at $x = 3$ ∴ a stationary point at $x = 3$ (QED)

Q6: Alg /5
Geom /8

Aw 1 for corr derivative
Aw 1 for solving $\frac{dy}{dx} = 0$

/2

ii) $\frac{d^2y}{dx^2} = 12x^2$

$12x^2 > 0$ at $x = 3$ ∴ the stationary point is a minimum

Aw 1 for correct 2nd deriv.
Aw 1 for correct reasoning

/2

iii) $\frac{d^2y}{dx^2} = 12x^2$

At $x = 0$

$\frac{d^2y}{dx^2} = 0$

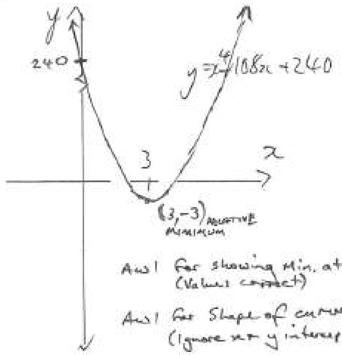
At $x = -1$ $\frac{d^2y}{dx^2} = 12(-1)^2$

$= 12 > 0$

At $x = 1$ $\frac{d^2y}{dx^2} = 12(1)^2$

$= 12 > 0$

No change in concavity has occured at $x = 0$ ∴ not an inflexion at (0, 240)



Aw1 for showing Min. at (3,-3) (Values correct)

Aw1 for Shape of curve (Ignore x or y intercepts)

Q31 KINGS

Consider the function $y = x^3 - 6x^2 + 9x + 2$.

(i) Find $y = f'(x)$? **1**

(ii) Find the y intercept. **1**

(iii) Find the coordinates of the two stationary points. **2**

(iv) Determine the nature of the stationary points. **2**

(v) Sketch the curve $y = f(x)$ for $0 \le x \le 5$ showing the stationary points. **3**

$f(x) = x^3 - 6x^2 + 9x + 2$

$f'(x) = 3x^2 - 12x + 9$

let $x = 0$ $y = 2$

$3x^2 - 12x + 9 = 0$

$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

$x = 1$ & 3 Aw1

when $x = 1$ $y = 1 - 6 + 9 + 2$

$= 6$

$x = 3$ $y = 27 - 54 + 27 + 2$

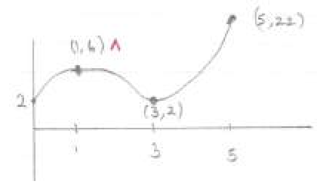
$y = 2$

∴ (1, 6) and (3, 2) Aw1

$f''(x) = 6x - 12$ Aw1

when $x = 1$ $y'' < 0$ ∴ max TP (1, 6) Aw1

$x = 3$ $y'' > 0$ ∴ min TP (3, 2)



Q32 **KINGS**

A function $f(x)$ is defined by $f(x) = 2x^3 - 3x^2$.

(i) Find all the roots of $f(x) = 0$. **1**
(ii) Find the turnings points for the curve $y = f(x)$ and determine their nature. **3**
(iii) Find the coordinates of the point of inflexion. **2**
(iv) Sketch the graph of $y = f(x)$, labelling the stationary points, points of inflexion and x-intercept. **2**
(v) For what values of x is the curve decreasing? **1**
(vi) For what values of x is the curve concave upwards? **1**

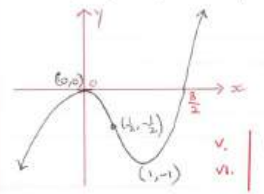
b i $f(x) = 2x^3 - 3x^2 = 0$
$x^2(2x-3) = 0$ $\therefore$ $x = 0$ and $x = 3/2$.

ii $f'(x) = 6x^2 - 6x$
$f'(x) = 0$
$0 = 6x(x-1)$ $\therefore$ $x = 0$ or $x = 1$ 1.
when $x = 1$ $f(1) = 2(1)^3 - 3(1)^2$
$= -1$
$x = 0$ $f(0) = 2(0)^3 - 3(0)^2$
$= 0$
$\therefore$ turning points @ $(0,0)$ & $(1,-1)$. 1.

$f''(x) = 12x - 6$
$f''(0) = -6$ $\therefore$ Max at $(0,0)$
$f''(1) = 6$ $\therefore$ Min at $(1,-1)$ 1.

iii Possible inflexion @ $f''(x) = 0$
$f''(x) = 12x - 6$
$0 = 12x - 6$ $\therefore$ $x = 1/2$. 1.
$f(1/2) = 2(\frac{1}{2})^3 - 3(1/2)^2$
$= -1/2$
Check concavity
@ $x = 0.4$ $f''(0.4) = -1.2 < 0$ 1.
$x = 0.6$ $f''(0.6) = 1.2 > 0$
$\therefore$ $(1/2, -\frac{1}{2})$ is a pt of inflexion.

iv

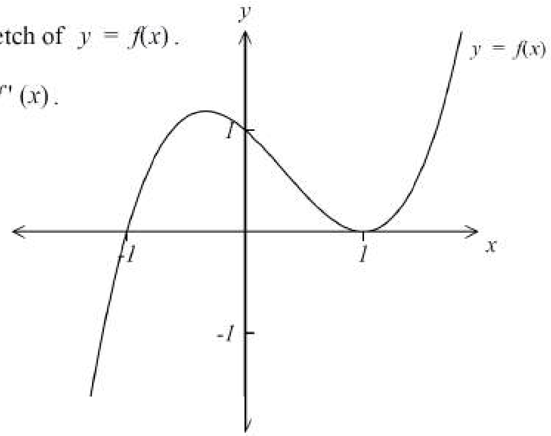


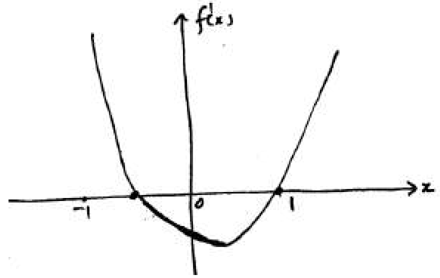
v. $0 < x < 1$
vi. $x > \frac{1}{2}$

Q33 **KINGS**

The diagram shows the sketch of $y = f(x)$.

Sketch the graph of $y = f'(x)$.





Q34 KINGS

Consider the curve with equation $y = 4x^4 + 8x^3 + 1$.

i. Find the stationary points and determine their nature. **4**

ii. Show that the point (-1, -3) is a point of inflexion. **2**

iii. Sketch the curve, labelling the stationary points, points of inflexion and y-intercept. **3**

$y' = 16x^3 + 24x^2$

$y' = 0$ for a $S.P.$

$8x^2(2x + 3) = 0$

$x = 0$ and $x = -\frac{3}{2}$

$y = 1$ and $y = -\frac{23}{4}$ respectively

Using a slope table

x	-2	-1.5	-1	0	1
y'	-32	0	8	0	40
slope	\	-	/	-	/

$\left(-\frac{3}{2}, -\frac{23}{4}\right)$ is a minimum and "(0,1) a horizontal point of inflexion

Or use of y'' to show concavity change

$y'' = 0$ is not sufficient for a point of inflection

ii)

$y'' = 48x^2 + 48x$

$y'' = 0$ for possible points of inflexion

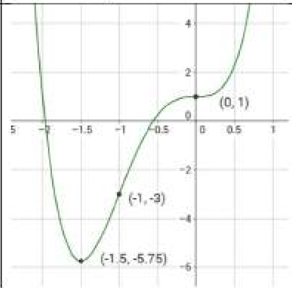
$48x(x + 1) = 0$

$x = -1, 0$

or sub x=1 into 2nd derivative

this point of inflection is NOT a horizontal point of inflection

iii)



Q35 KINGS

Consider the curve $y = x^3 + 3x^2 - 9x - 11$.

(i) Find $\frac{dy}{dx}$. **1**

(ii) Find the coordinates of the two stationary points. **2**

(iii) Determine the nature of the stationary points. **2**

(iv) Sketch the curve for the domain $-4 \le x \le 3$. **2**

(v) By drawing an appropriate line on your graph, or otherwise, solve $x^3 + 3x^2 - 9x - 27 = 0$. **1**

(i) $y = x^3 + 3x^2 - 9x - 11$

$\frac{dy}{dx} = 3x^2 + 6x - 9 = 3(x^2 + 2x - 3)$

(ii) $\frac{dy}{dx} = 0$

$\therefore 3x^2 + 6x - 9 = 0$

$3(x^2 + 2x - 3) = 0$

$3(x-1)(x+3) = 0$

$\therefore x = 1$ or $x = -3$

$(-3, 16)$ and $(1, -16)$

(iii) $\frac{d^2y}{dx^2} = 6x + 6 = 6(x+1)$

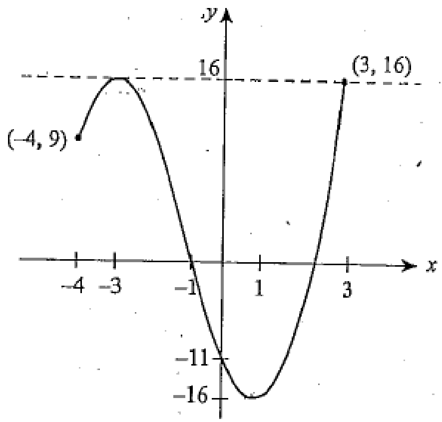
$\therefore f''(-3) = -12$

i.e. $(-3, 16)$ is a maximum

and $f''(1) = 12$

i.e. $(1, -16)$ is a minimum

(iv)



(v) $x^3 + 3x^2 - 9x - 27 = 0$

$\therefore x^3 + 3x^2 - 9x - 11 - 16 = 0$

$x^3 + 3x^2 - 9x - 11 = 16$

$\therefore$ drawing $y = 16$ will provide solutions

i.e. $x = -3, 3$

Q36 NEAP

Consider the function $f(x) = 1 - 3x + x^3$, in the domain $-2 \le x \le 3$.

(i) Find the coordinates of the turning points and determine their nature. **3**

(ii) Draw a sketch of the curve $y = f(x)$ in the domain $-2 \le x \le 3$, clearly showing all its essential features. **2**

(iii) What is the maximum value of the function $f(x)$ in the domain $-2 \le x \le 3$? **1**

(a) (i) $f'(x) = -3 + 3x^2,\ f''(x) = 6x$

For turning points $f'(x) = 0$,

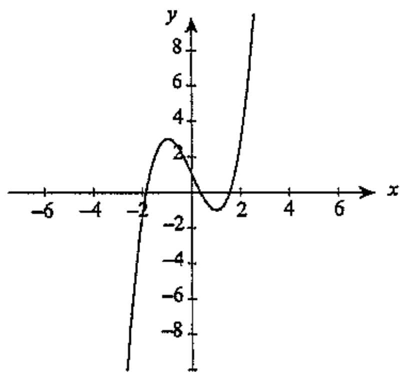
hence $-3 + 3x^2 = 0,\ x^2 = 1,\ x = \pm 1$

Turning points are $(1, -1)$ and $(-1, 3)$

Nature of these points: $f''(1) = 6,\ f''(-1) = -6$

Hence, $(1, -1)$ is a minimum and $(-1, 3)$ is a maximum.

(ii)



(iii) $\max f(x) = 19$, when $x = 3$.

Q37 NORMANHURST

A function is defined by $f(x) = 2x^3 - 6x + 3$.

i. Find the coordinates of the turning points of the graph $y = f(x)$ and determine their nature. **3**

ii. Hence sketch the graph of $y = f(x)$, showing the turning points and the y intercept. **2**

(3 marks)

$$f(x) = 2x^3 - 6x + 3$$
$$f'(x) = 6x^2 - 6$$

Stationary pts occur when $f'(x) = 0$:

$$6(x^2 - 1) = 0$$
$$(x - 1)(x + 1) = 0$$
$$\therefore x = \pm 1$$
$$f''(x) = 12x$$

- When $x = -1$:

$$f(-1) = -2 + 6 + 3 = 7$$
$$f''(-1) = -12 < 0$$

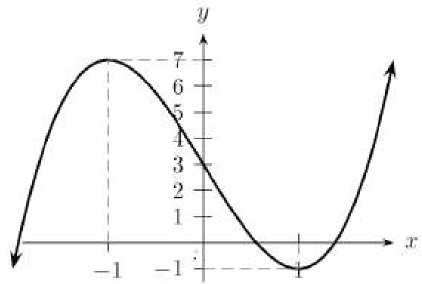
Hence $(-1, 7)$ is a local maximum.

- When $x = 1$:

$$f(1) = 2 - 6 + 3 = -1$$
$$f''(1) = 12 > 0$$

Hence $(1, 1)$ is a local minimum.

ii. (2 marks)



Q38 **NSBHS**

A function is defined by $f(x) = 2x^3 - 6x + 3$.

i. Find the coordinates of the turning points of the graph $y = f(x)$ and determine their nature. **2**

ii. Hence sketch the graph of $y = f(x)$, showing the turning points and the y intercept. **2**

d) $f(x) = 2x^3 - 6x + 3$

(i) $f'(x) = 6x^2 - 6$

$f''(x) = 12x$

Stat Pts: $f'(x) = 0$

$6x^2 - 6 = 0$

$6x^2 = 6$

$x^2 = 1$

$x = 1, -1$

$y = -1, 7$

Test $(1,-1)$

$f''(x) = 12$

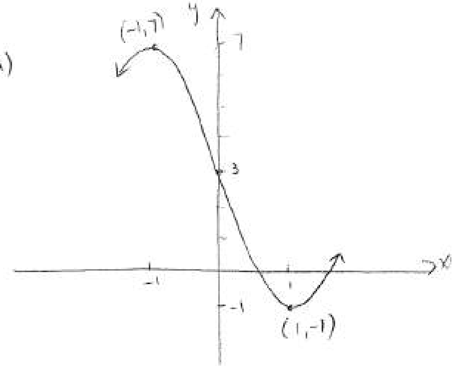
> 0 $\therefore \boxed{\min (1,-1)}$ 1

Test $(-1,7)$

$f''(x) = -12$

< 0 $\therefore \boxed{\max -1,7}$ 1

(ii)



Q39 **NSBHS**

Given the equation $y = -x(x-5)^2 = -x^3 + 10x^2 - 25x$

i) Find any stationary points and determine their nature. 2

ii) Find any points of inflexion. 2

iii) Hence sketch the curve, clearly indicating the intercepts, stationary points and points of inflexion. 4

Q14. (10)

a) $y = -x(x-5)^2 = -x^3 + 10x^2 - 25x$

i) $y' = -3x^2 + 20x - 25$

$= -(3x-5)(x-5)$

Stationary points when $y' = 0$

i.e. at $x = \frac{5}{3}$ and $x = 5$ \1

Test points:

$y'' = -6x + 20$

At $x = 5/3$,

$y'' = -6\left(\frac{5}{3}\right) + 20$

$= 10$

> 0

Hence local minimum

$y = \left(-\frac{5}{3}\right)\left(\frac{5}{3} - 5\right)^2$

$= \left(-\frac{5}{3}\right)\left(\frac{15}{3}\right)^2$

$= -\frac{500}{27} \simeq 18.5$

$\left(\frac{5}{3}, -\frac{500}{27}\right)$ local min. \1

At $x = 5$,

$y'' = -6(5) + 20$

$= -10$

< 0

Local max

$y = (-5)(5-5)^2 = 0$

$(5,0)$ local max. \1

Deduct 1 mark for not testing

(ii) Inflexion possible at

$y'' = 0$

i.e. $0 = -6x + 20$

$x = \frac{10}{3}$ \1

wrong! Need to check both sides for concavity changes.

$y' = -3\left(\frac{100}{9}\right) + 20\left(\frac{10}{3}\right) - 25$

$\neq 0$

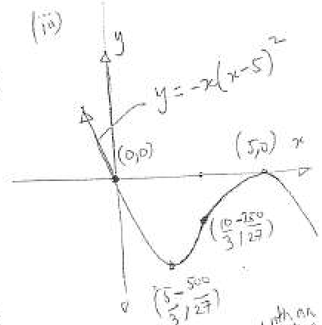
$\therefore$ Point of inflexion. \1 check

$y = \left(-\frac{10}{3}\right)\left(\frac{10}{3} - 5\right)^2$

$= \left(-\frac{10}{3}\right)\left(-\frac{5}{3}\right)^2$

$= -\frac{250}{27} \approx -9.26$

(iii)



\1 correct shape with an inflexion

\1 stationary points consistent with (i)

... labelling ...

Q40 NSBHS

A function is defined by $f(x) = x^3 - 3x^2 - 9x + 22$.

i. Find the coordinates of the turning points of the graph $y = f(x)$ and determine their nature. **3**

ii. Find the coordinates of the point(s) of inflexion. **2**

iii. Hence sketch the graph of $y = f(x)$, showing the turning points, the point(s) of inflexion and the y intercept. **2**

i. (3 marks)

$$\begin{aligned} y &= x^3 - 3x^2 - 5x + 22 \\ y' &= 3x^2 - 6x - 9 \\ &= 3\left(x^2 - 2x - 3\right) \\ &= 3(x-3)(x+1) \end{aligned}$$

Stationary points occur when $y' = 0$:

$$\therefore x = -1, 3$$

When $x = -1$,

$$y = (-1)^3 - 3(-1)^2 - 9(-1) + 22 = 27$$

When $x = 3$,

$$y = 3^3 - 3(3^2) - 9(3) + 22 = -5$$

Finding the second derivative,

$$\begin{aligned} y' &= 3x^2 - 6x - 9 \\ y'' &= 6x - 6 \end{aligned}$$

When $x = -1$,

$$y' = 6(-1) - 6 < 0$$

$\therefore$ $(-1, 27)$ is a local max. When $x = 3$,

$$y' = 6(3) - 6 > 0$$

$\therefore (3, -5)$ is a local min.

ii. (2 marks)

$$\begin{aligned} y' &= 3x^2 - 6x - 9 \\ y'' &= 6x - 6 \end{aligned}$$

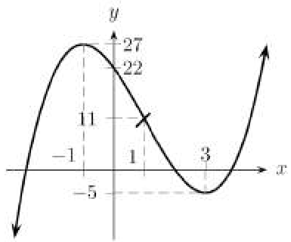
Pt of inflexion occurs when $y'' = 0$:

$$\begin{aligned} 6x - 6 &= 0 \\ x &= 1 \end{aligned}$$

x	0	1	2
$\frac{d^2y}{dx^2}$	$-$ -6	0	$+$ 6
y	$\frown$		$\smile$

When $x = 1$, $y = 1 - 3 - 9 + 22 = 11$. Hence $(1, 11)$ is a point of inflexion as a change of sign of the 2nd derivative also occurs.

iii. (2 marks)



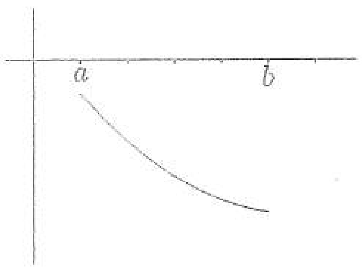
Q1 HURLSTONE

A continuous curve, $y = f(x)$ has the following properties for the interval $a \le x \le b$. 2

$$f(x) < 0,\ f'(x) < 0,\ f''(x) > 0$$

Sketch a curve satisfying these conditions for $a \le x \le b$.

(e) $f(x) < 0$: y values negative
$f'(x) < 0$: decreasing
$f''(x) > 0$: concave up



2 marks : correct solution

1 mark : displays graph with two of the given properties correct

Q 2 ACE

What values of x is the curve $f(x) = x^3 + x^2$ concave down?

(A) $x < -\frac{1}{3}$

(B) $x > -\frac{1}{3}$

(C) $x < -3$

(D) $x > 3$

Concave down when $f''(x) < 0$

$f'(x) = 3x^2 + 2x$

$f''(x) = 6x + 2$

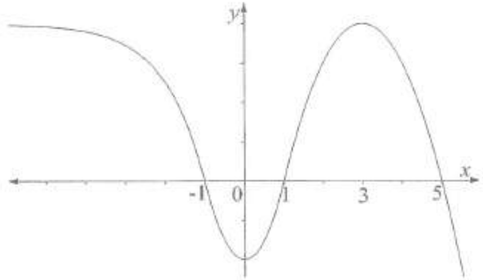
$6x + 2 < 0$

$x < -\frac{1}{3}$

1 Mark: A

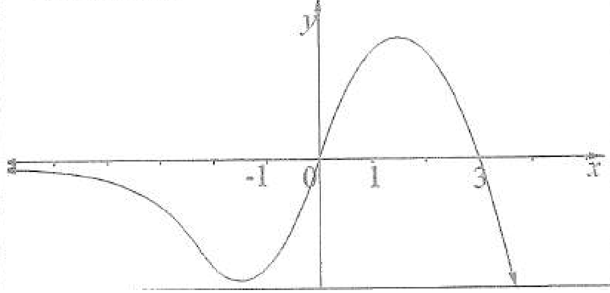
Q3 HURLSTONE

2



Sketch the derivative function of the curve above.

(f) At stationary points derivative $= 0$. $\therefore y' = 0$ at $x = 0, 3$.
Curve decreasing when $x < 0$ and $x > 3$, and increasing when $0 < x < 3$.
$\therefore y'$ is negative when $x < 0$ and $x > 3$, and y' positive when $0 < x < 3$.
Maximum, minimum values of y' occur at points of inflexion on y.



2 marks : correct solution

1 mark : substantial progress towards correct solution

Q4 GOSFORD

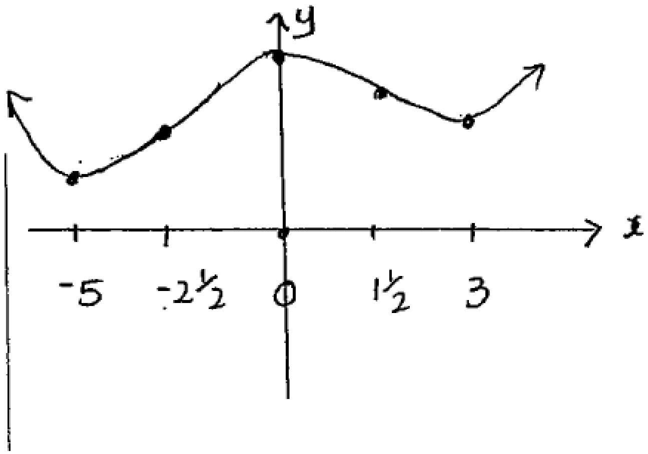
Give a possible sketch for a curve which has the following defining features: **2**

When x = -5, 0 and 3,

$\frac{dy}{dx} = 0$.

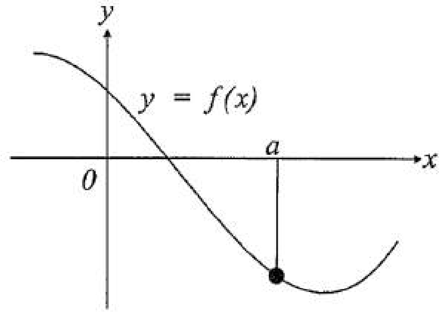
Also $\frac{d^2y}{dx^2} > 0$ for x < -2½ and x > 1½, *while* $\frac{d^2y}{dx^2} < 0$ for - 2½ < x < 1½.

d)



Q5 GIRRAWEEN

The diagram shows the graph of $y = f(x)$.



Which of the following statements is true?

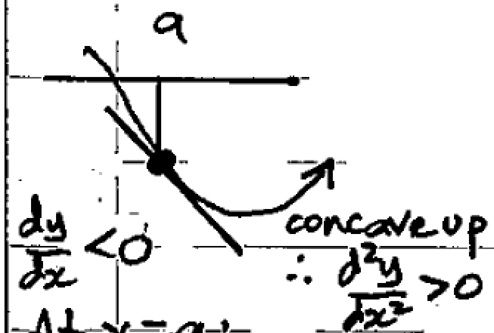
(A) $f'(a) < 0$ and $f''(a) < 0$

(B) $f'(a) < 0$ and $f''(a) > 0$

(C) $f'(a) > 0$ and $f''(a) < 0$

(D) $f'(a) > 0$ and $f''(a) > 0$

6)



At $x = a$:

* gradient negative

$\therefore f'(a) < 0$

* concave UP

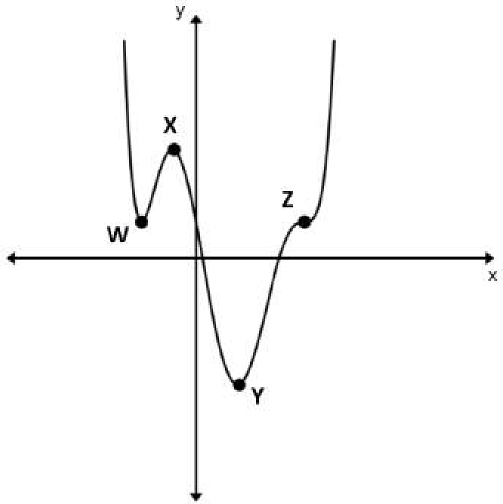
$\therefore f''(a) > 0$

At $x = a$:

$\therefore f'(a) < 0$ and $f''(a) > 0$ B

Q6 CARINGBAH

The diagram below shows the graph of $y = f(x)$. Points W, X, Y and Z are stationary points and Z is a point of inflexion on $f(x)$.



Which of the points on $y = f(x)$ corresponds to the description:

$$y > 0 \qquad \frac{dy}{dx} = 0 \qquad \frac{d^2y}{dx^2} < 0$$

A. W

$\boxed{\text{B. X}}$

C. Y

D. Z

Q7 BAULKHAM

A function $y = f(x)$ is defined by the following features:

$f''(x) > 0$ for $x < -1$ and $1 < x < 3$

$f'(x) = 0$ when $x = -3, 1$ and 5

$f(x) = 0$ when $x = 1$

(i) Identify the x values of the stationary points and determine the nature of each point. **2**

(ii) Sketch a possible graph of the function. **1**

(i) as $f'(x) = 0$, -3, 1 and 5 are x values of stationary points

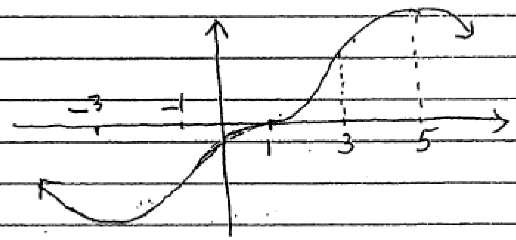
$f''(x) > 0$ for $x < -1$ $\Longrightarrow$ Concave up

$f''(x) > 0$ for $1 < x < 3$ $\Longrightarrow$ Concave up.

$\therefore$ at $x = -3$ a minimum turning point

$x = 5$ a maximum turning point

all ✓



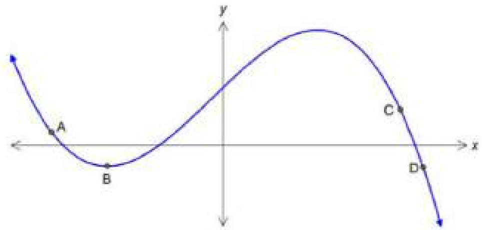
✓

Q8

State which point on the diagram relates to the following

$y > 0,\ \frac{dy}{dx} < 0,\ \frac{d^2y}{dx^2} > 0.$

BAULKHAM

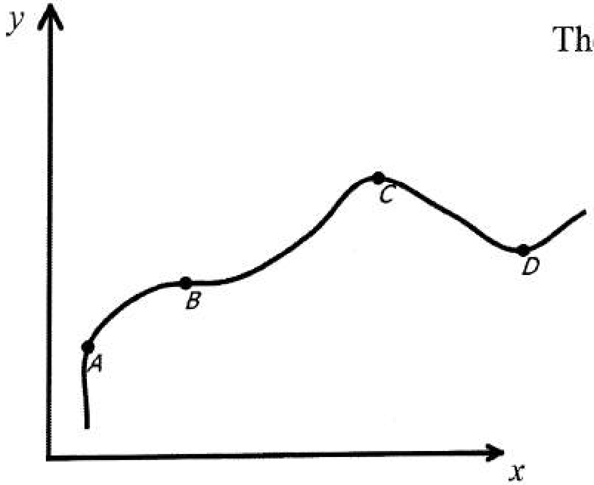


$\boxed{\text{(A) } A}$ (B) B (C) C (D) D

A point on the graph of $y = f(x)$ has $f'(x) = 0$ *and* $f''(x) = 0$.

Q9

BAULKHAM



The point is

(A) A

$\boxed{\text{(B) } B}$

(C) C

(D) D

Q10

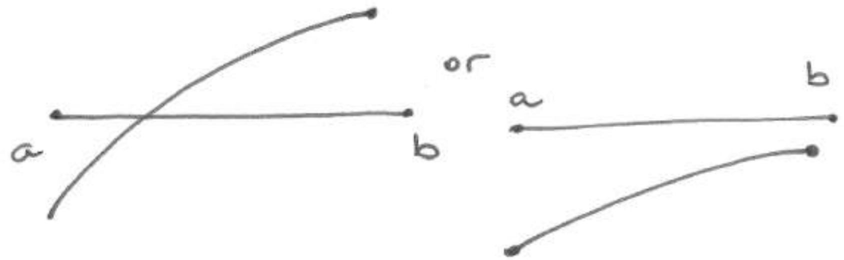
BARKER

Over the interval $a \le x \le b$, a curve $y = f(x)$ has the following properties: **2**

$f(a) < 0\ ,\ f'(x) > 0$ and $f''(x) < 0$

Draw this section of the curve $y = f(x)$ illustrating all of the above information.

Q15b)



Q11 **BARKER**

The curve $y = f(x)$ has two stationary points at $x = 1$ and $x = a$ **2**

If $f''(x) = 6x - 2$, find the value of a.

a) $f''(x) = 6x - 2$

$f'(x) = 3x^2 - 2x + C$

$\frac{dy}{dx} = 0 \quad 0 = 3 - 2 + C$

$C = -1$

$f'(a) = 0 = 3a^2 - 2a - 1$

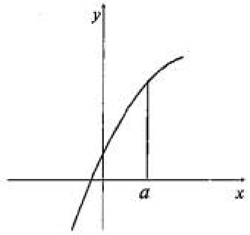
$(3a+1)(a-1) = 0$

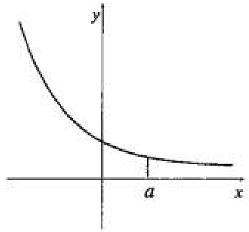
$a = -\frac{1}{3},\ 1$

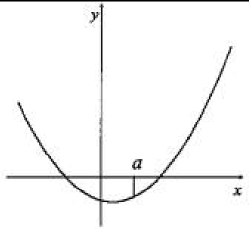
$\therefore a = -\frac{1}{3}$

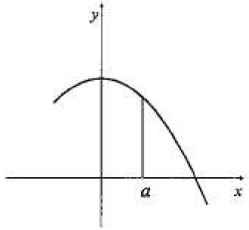
Q12 BAUL

Which diagram indicates $f'(a)>0$ and $f''(a)<0$?

A


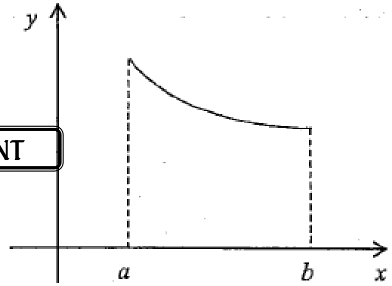
B


C


D


Q13 INDEPENDENT

For the function $y=f(x)$, $a<x<b$ graphed below:



which of the following is true?

A $f'(x)>0$ and $f''(x)>0$

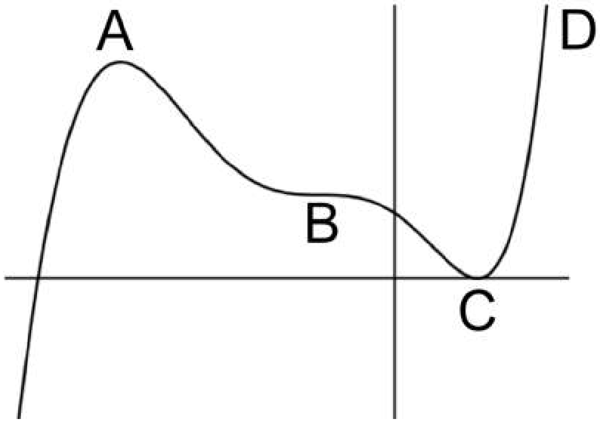
B $f'(x)>0$ and $f''(x)<0$

$\boxed{\text{C} \quad f'(x)<0 \text{ and } f''(x)>0}$

D $f'(x)<0$ and $f''(x)<0$

Q14 KILARA

At which point on the graph of $f(x)$ shown below is $f'(x)<0$ and $f''(x)=0$?



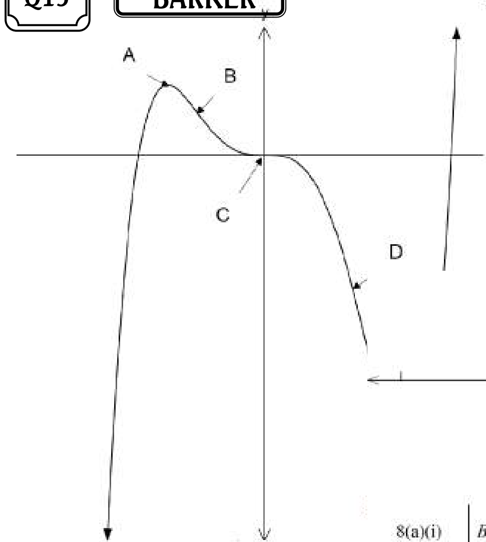
(A) A

$\boxed{\text{(B)} \quad \text{B}}$

(C) C

(D) D

Q15 BARKER

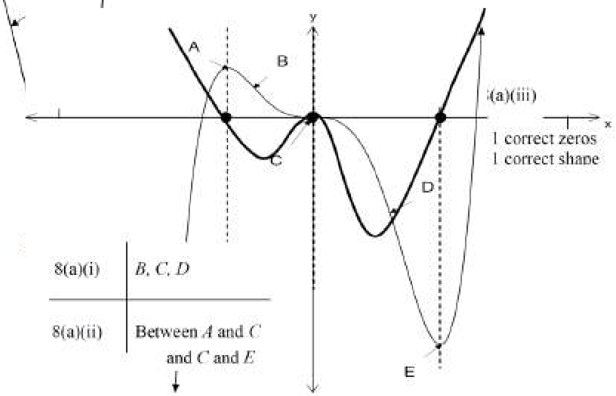


The graph of the curve $y = f(x)$ is drawn below.

(i) Name the points of inflexion. **1**

(ii) When is the graph decreasing? **1**

(iii) Sketch the gradient function. **2**

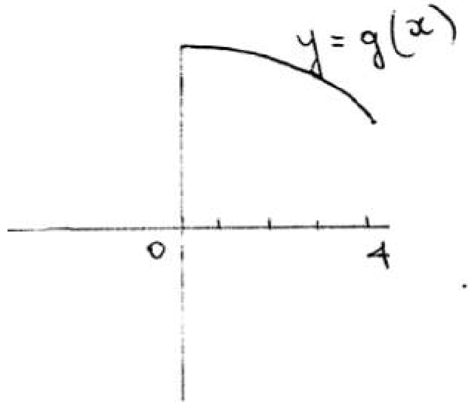


8(a)(i)	B, C, D
8(a)(ii)	Between A and C and C and E

Q16

For all x in the domain $0 \le x \le 4$, a function $g(x)$ satisfies $g'(x) < 0$ and $g''(x) < 0$
Sketch a possible graph of $y = g(x)$ in this domain. **3**

KINGS

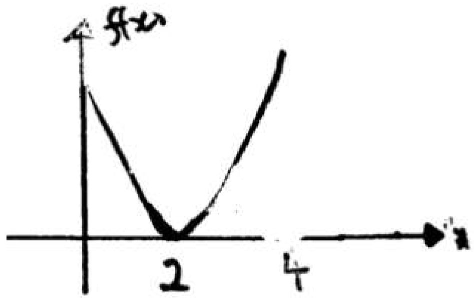


Q17

KINGS

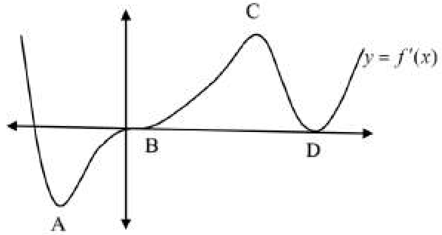
Sketch as much of the continuous function $f(x)$ as is indicated by the data

$f(2) = 0$, $f^1(x) < 0$ for $0 < x < 2$ and $f^1(x) > 0$ for $2 < x < 4$ **2**



Q18 **KNOX**

The diagram shows a sketch of the gradient function $y = f'(x)$ passing through the points A, B, C and D.



Which point represents the horizontal point of inflexion of the curve $y = f(x)$?

(A) Point A
(B) Point B
(C) Point C
(D) Point D

Q19 **BARKER**

Which of the following statements is true if $(2, -5)$ is a minimum turning point of $f(x)$ and $f(x) = -f(-x)$?

(A) $(-2, -5)$ is a maximum turning point of $f(x)$

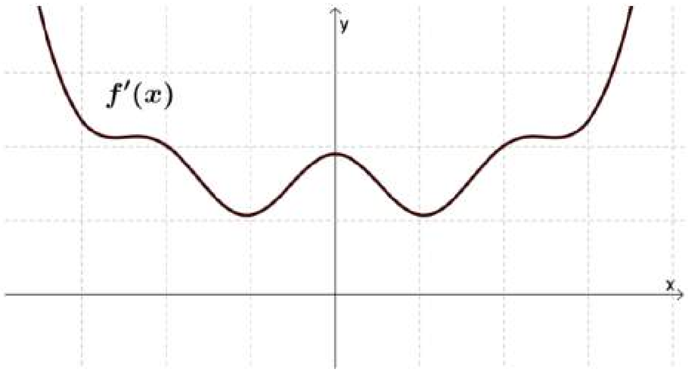
(B) $(-2, 5)$ is a minimum turning point of $f(x)$

(C) $(-2, -5)$ is a minimum turning point of $f(x)$

(D) $(-2, 5)$ is a maximum turning point of $f(x)$

Q20

NSBHS



Looking at the graph of $y = f'(x)$ above, which of the following is true?

(A) $f(x)$ is an odd function with a local maximum at $f(0)$.

(B) $f(x)$ is an odd function with a point of inflexion at $f(0)$.

(C) $f(x)$ is an even function with a local maximum at $f(0)$.

(D) $f(x)$ is an even function with a point of inflexion at $f(0)$.

~~Grow~~ $f(x)$ is always > 0, so always increasing.
$\to$ no turning points.
$f''(0) = 0$, $f'(x)$ not changing sign $\to$ inflexion
(B) odd function with inflexion at $f(0)$

Q1 INDEPENDENT

The gradient function of a curve $y=f(x)$ is given by $f'(x)=4+\frac{1}{x^2}$. **3**

The curve passes through the point $(1,2)$. Find the equation of the curve.

Answer

$f'(x)=4+x^{-2}$ $\quad \therefore f(x)=4x-x^{-1}+c$. But $f(1)=2 \Rightarrow 4-1+c=2 \quad \therefore c=-1$

Hence curve has equation $y=4x-\frac{1}{x}-1$

Q2 KINGS

Find the equation of the curve $y=f(x)$, given that $\frac{d^2y}{dx^2}=2x+3$ **3**

and that there is a minimum turning point at (1, 3)

5. (a) $\frac{d^2y}{dx^2}=2x+3$

$\frac{dy}{dx}=x^2+3x+c$

at $(1,3)$ $\frac{dy}{dx}=0$

$1^2+3+c=0$

$c=-4$

So $\frac{dy}{dx}=x^2+3x-4$

$y=\frac{x^3}{3}+\frac{3x^2}{2}-4x+c$

at $(1,3)$

$3=\frac{1}{3}+\frac{3}{2}-4+c$

$c=5\frac{1}{6}$

$\therefore y=\frac{x^3}{3}+\frac{3x^2}{2}-4x+5\frac{1}{6}$

Q3 INDEPENDENT

The gradient function of a curve is $\frac{dy}{dx}=3-\frac{2}{x^2}$. Find the equation of the curve if it passes through the point $(1,-2)$. **2**

b)

$\frac{dy}{dx}=3-\frac{2}{x^2}$

$\frac{dy}{dx}=3-2x^{-2}$

$y=3x+2x^{-1}$

$y=3x+\frac{2}{x}$

$-2=3+\frac{2}{1}+c$

$-7=c$

$y=3x+\frac{2}{x}-7$

Q4 KINGS

For the function $y=f(x)$, where $f'(x)=2x-2$, if there is a stationary point at (1, 3), then $f(x)$ equals

(A) x^2-2x

(B) 2

(C) x^2-2x+4

(D) x^2-2x-4

$f'(x)=2x-2$

$f(x)=x^2-2x+c$

subst $(1,3)$

$3=1-2+c$

$c=4$

$f(x)=x^2-2x+4$

$\boxed{C}$

Q5 KINGS

A function $y = f(x)$ has its gradient function given by $\frac{dy}{dx} = 12x^2 - \frac{3}{x^2} + k$ where k is a constant.

If the gradient of the tangent at the point (1, 5) on the curve is 9

(i) evaluate k **1**

(ii) find the equation of the function $y = f(x)$ **2**

i) $\frac{dy}{dx} = 12x^2 - 3x^{-2} + k$

When $x=1$ $\frac{dy}{dx} = 9$

$12 - 3 + k = 9$

$k = 0$

Aw 1 for $k=0$, with working

1

ii) $y = 4x^3 - \frac{3x^{-1}}{-1} + c$

$y = 4x^3 + \frac{3}{x} + c$

Aw 1 for Primitive, with const.
Aw 1 for Correct Ans

When $x=1$ $y=5$

$5 = 4 + 3 + c$

$c = -2$

The function $y = f(x) = 4x^3 + \frac{3}{x} - 2$

2

[Accept $y = 4x^3 + 3x^{-1} - 2$]

Q6 KINGS

The curve $y = f(x)$ passes through the point $\left(1, \frac{2}{3}\right)$ and $f'(x) = \frac{1}{2}(x - 3)^2$.

Find the equation of the curve. **3**

$y = f(x)$ $(1, 2/3)$

$f'(x) = \frac{1}{2}(x-3)^2$

$f(x) = \frac{(x-3)^3}{6} + c$ Aw 1

$f(1) = 2/3$

$\therefore \frac{(-2)^3}{6} + c = 2/3$ Aw 1

$-\frac{8}{6} + c = 2/3 + 4/3$

$= 2$

$\therefore f(x) = \frac{(x-3)^3}{6} + 2$ Aw 1

Q7 HURLSTONE

Which of the following conditions for $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ describe the slowing growth of a variable P?

(A) $\frac{dP}{dt}>0$ and $\frac{d^2P}{dt^2}>0$

(B) $\frac{dP}{dt}<0$ and $\frac{d^2P}{dt^2}<0$

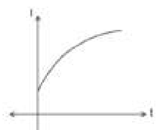
(C) $\frac{dP}{dt}>0$ and $\frac{d^2P}{dt^2}<0$

(D) $\frac{dP}{dt}<0$ and $\frac{d^2P}{dt^2}>0$

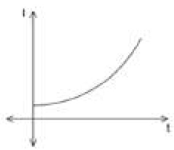
Q8 BAULKHAM

Interest rates are increasing at an decreasing rate . Which of the following graphs represents the above statement

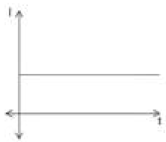
(A)



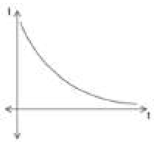
(B)



(C)



(D)



Q9 NSBHS

If M is decreasing at an increasing rate, what does this suggest about $\frac{dM}{dt}$ and $\frac{d^2M}{dt^2}$?

(A) $\frac{dM}{dt}<0$ and $\frac{d^2M}{dt^2}<0$

(C) $\frac{dM}{dt}<0$ and $\frac{d^2M}{dt^2}>0$

(B) $\frac{dM}{dt}>0$ and $\frac{d^2M}{dt^2}<0$

(D) $\frac{dM}{dt}>0$ and $\frac{d^2M}{dt^2}>0$

Q10 NSBHS

Which of the following conditions for $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ describe the slowing growth of a variable P? **1**

(A) $\frac{dP}{dt}>0$ and $\frac{d^2P}{dt}>0$.

(C) $\frac{dP}{dt}>0$ and $\frac{d^2P}{dt}<0$.

(B) $\frac{dP}{dt}<0$ and $\frac{d^2P}{dt}<0$.

(D) $\frac{dP}{dt}<0$ and $\frac{d^2P}{dt}>0$.

Q11 NSBHS

Find $f(x)$ if $f'(x)=2x+\frac{1}{x^2}$ and the curve passes through the point $(1,2)$. **3**

$$f(x)=\int\left(2x+x^{-2}\right)\,dx=x^2-x^{-1}+C$$
$$f(1)=1-1^{-1}+C=2$$
$$\therefore C=2$$
$$\therefore f(x)=x^2-\frac{1}{x}+2$$